Tutorial 1:
Regular Expression and Finite Automata

Exercise 1:

- Give this Regex Over the alphabet {0,1,2,3,4,5,6,7,8,9} , d=(0-9)

1. Regular expression that denotes numbers of only three digits :
2. Regular expression that denotes numbers that could be divided by 5 :
3. Regular expression that denotes numbers that starts with 4 and ends with 7:

Exercise 2:

For each of the languages below, explain the language and draw an automaton that recognizes it using a construction method.

1. L is the language denoted by (aba)|(bab).
2. L is the language denoted by (aba)*| (bab)*.
3. L = {u ∈{a, b}* such that u contains the factor bbb}.
4. L = {u ∈{a, b}* such that u does not contain the bbb factor

Exercice 3 :

We consider the following regular expression: (a|b)*abb

Give the Finite automata of this expression.

Exercise 4:

Give the RegEx and the automata corresponding to these languages on the alphabet {a,b}

- The language of words containing at least once the letter a
- The language of words containing at most once the letter a:
- The language of words containing an even number of times the letter a

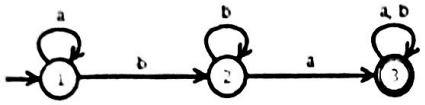
Exercise 5:

We consider the following transition table, so give the corresponding Automata.

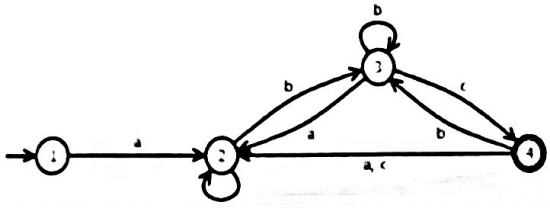
	a	b
0	1	0
1	0	2
2	3	0
3	1	2

Exercise 6: Give the languages corresponding to each automaton

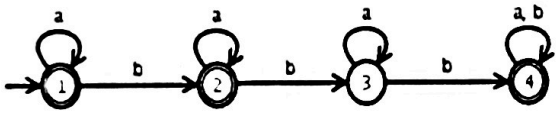
a)



b)



c)



d)

